4. LOSS OF INDEPENDENCE

35. On the effects of drugs on time perception, see R. A. Sewell et al., “Acute Effects of THC on Time Perception in Frequent and Infrequent Cannabis Users,” *Psychopharmacology* 226 (2013): 401–13; the direct experience is astonishing.

36. Valtteri Arstila, “Time Slows Down during Accidents,” *Frontiers in Psychology* 3 (2012): 196.

37. In our cultures. There are others with a profoundly different notion of time: D. L. Everett, *Don’t Sleep, There are Snakes*, (New York: Pantheon, 2008).

38. Matthew 20:1–16.

39. Peter Galison, *Einstein’s Clocks, Poincaré’s Maps* (New York: Norton, 2003), p. 126.

40. An excellent panoramic history of the way in which technology has progressively modified our concept of time can be found in Adam Frank, *About Time: Cosmology and Culture at the Twilight of the Big Bang* (New York: Free Press, 2001).

41. D. A. Golombek, I. L. Bussi, and P. V. Agostino, “Minutes, Days and Years: Molecular Interactions among Different Scales of Biological Timing,” *Philosophical Transactions of the Royal Society. Series B: Biological Sciences* 369 (2014).

42. Time is: “number of change, with regard to before and after” (Aristotle, *Physics* IV.219b2; see also 232b22–3).

43. Aristotle, *Physics*, trans. Robin Waterfield, with an introduction and notes by David Bostock (Oxford: Oxford University Press, 1999), p. 105.

44. Isaac Newton, *Philosophiae Naturalis Principia Mathematica*, Book I, def. VIII, scholium.

45. Ibid.

46. An introduction to the philosophy of space and of time can be found in B. C. van Fraassen, *An Introduction to the Philosophy of Time and Space* (New York: Random House, 1970).

47. Newton’s fundamental equation is $F = m\, d^2x/dt^2$. Note that time t is squared: this reflects the fact that the equation does not distinguish t from $-t$, that is to say, it is the same backward or forward in time, as I explain in chapter 2.

48. Curiously, many contemporary manuals of the history of science present the discussion between Leibniz and the Newtonians as if Leibniz were the heterodox figure with audacious and innovative relationist ideas. In reality, the opposite was the case: Leibniz defended (with a new wealth of arguments) the dominant traditional understanding of space, which from Aristotle to Descartes had always been relationist.

49. Aristotle’s definition is more precise: the place of a thing is the *inner boundary* of that which surrounds the thing, an elegant and rigorous definition.

5. QUANTA OF TIME

50. I speak of this in more depth in *Reality Is Not What It Seems*, trans. Simon Carnell and Erica Segre (New York: Riverhead Books, 2017).

51. It is not possible to locate a degree of liberty in a region of its phase space within a volume smaller than the Planck constant.

52. The speed of light, the Newton constant and the Planck constant.

53. Maimonides, *The Guide for the Perplexed* I.73.106a.